SMART PARKING SYSTEM USING IoT AND MACHINE LEARNING

Phase 1 – Project Proposal

Course	ML 402331 – IoT Machine Learning Project
Project Track	Hardware Track
Project Type	Smart Parking System

1. Project Idea and User Benefit

This project proposes a low-cost Smart Parking System for a small parking area with a capacity of four vehicles. The intended user is a driver approaching the parking entrance. The system will detect vehicle entry and exit, estimate the number of occupied spaces, and display whether the parking area is Available or Full. Its intended benefit is to help drivers make a quicker parking decision and reduce unnecessary searching.

2. Problem Statement

Drivers may spend unnecessary time searching for available parking spaces, particularly in busy areas. This can contribute to local congestion, wasted fuel, and inconvenience. A small, affordable system that detects vehicle movement and displays current parking availability may provide a practical way to communicate whether a parking area has remaining capacity.

3. Measurable Research Question

How accurately can a smart parking prototype using an ESP32 and two sequential IR sensors classify the parking area as Available or Full from recorded sensor-event data?

The question will be evaluated using a labeled dataset collected from the prototype. Evaluation measures will include accuracy, precision, recall, F1-score, and a confusion matrix on held-out test data. Results will be compared with the system's simple occupancy-counting rule.

4. Proposed System and Hardware Operation

Two IR sensors will be installed sequentially at the entrance. The order of activation will be used to infer whether a vehicle is entering or exiting. The ESP32 will update an occupancy counter in the range 0–4. An I2C LCD will display occupied and available spaces, and an SG90 servo motor will demonstrate a barrier arm.

• ESP32 development board – reads sensor outputs and runs the control logic.

• Two IR sensors – detect vehicle passage and direction from activation order.

• SG90 servo motor – demonstrates opening and closing the entrance barrier.

• I2C LCD – displays occupancy and parking availability.

Limitation: Two entrance sensors estimate total occupancy by counting entry and exit events; they do not detect each individual parking bay. Debounce timing, valid-sequence checks, and occupancy limits will be used to reduce duplicate or invalid counts.

5. Machine Learning Task, Target, and Features

Task: Binary Classification. **Target:** Parking_Status. **Classes:** Available (0) when verified occupancy is 0–3; Full (1) when verified occupancy is 4.

Planned features: IR1 state, IR2 state, detected event direction (entry/exit/none), time interval between sensor activations, and entry/exit event counts within a defined time window. Timestamp-derived features may include hour or test-session identifier.

The target will be assigned from verified occupancy. Features that directly reveal the target or are calculated from it will be excluded to reduce data leakage. A rule-based occupancy counter will be retained as a baseline. Candidate classifiers include Logistic Regression, Decision Tree/Random Forest, and SVM.

6. Data Collection Plan

The ESP32 will record timestamped sensor events during controlled vehicle-entry and exit trials. Each trial will be checked against the known movement direction and current verified occupancy. Records will be stored as CSV. The project target is at least 500 labeled records.

• Planned columns: timestamp, IR1 state, IR2 state, detected event, sensor activation interval, verified occupancy, and Parking_Status.

• Labels: Available for occupancy 0–3; Full for occupancy 4.

• Collection: repeat entry and exit trials across different occupancy levels and repeated runs.

• Quality control: debounce signals, reject invalid sequences, keep occupancy between 0 and 4, check missing values, and manually verify a sample of labels.

Data will be split into training and test subsets before model fitting. Evaluation will use accuracy, precision, recall, F1-score, and a confusion matrix. The actual record count and collection limitations will be reported; this proposal does not claim that the 500 records have already been collected.

7. Tools and Software Stack

Tool	Purpose / justification	Current status
Arduino IDE	Develop and upload ESP32 firmware.	Selected; installation/setup to confirm
Python + Google Colab	Prepare data, train models, and evaluate results.	Notebook created and run
Pandas / NumPy	Load, clean, and transform sensor-event data.	Used in Colab
scikit-learn	Train and evaluate ML models; used for Phase 1 Linear Regression exercise.	Used in Colab
Matplotlib	Visualize data and model results.	Used in Colab
GitHub	Store documentation, notebook, firmware, and dataset.	Repository created

8. Bill of Materials (Hardware Track)

Component	Qty	Purpose	Estimated cost (JOD)
ESP32 development board	1	Main controller / data logger	6–10
IR sensor modules	2	Detect vehicle passage direction	2–4 total
SG90 servo motor	1	Barrier demonstration	2–3
I2C LCD	1	Display occupancy and status	3–5
Breadboard	1	Prototype assembly	1–2
Jumper wires	As needed	Electrical connections	1–2
USB cable / suitable 5V supply	1	Power and programming	1–3
Estimated total			16–29

Costs are preliminary estimates and should be confirmed with local prices. The servo may require a separate regulated 5V supply with common ground. Check IR module output voltage before connecting it to ESP32 GPIO pins.

9. Related Works

These three works are included as preliminary related literature. Verify their full bibliographic metadata against publisher or library records before final submission.

1. Idris, M. Y. I., et al. (2009). “Car Park System: A Review of Smart Parking System and Its Technology.” Information Technology Journal, 8(2), 101–113.

2. Geng, Y., & Cassandras, C. G. (2013). “New Smart Parking System Based on Resource Allocation and Reservations.” IEEE Transactions on Intelligent Transportation Systems, 14(3), 1129–1139.

3. Khanna, A., & Anand, R. (2016). “IoT Based Smart Parking System.” In 2016 International Conference on Internet of Things and Applications (IOTA).

10. Phase 1 Public-Dataset and Linear-Regression Exercise

The starter notebook uses the public Diabetes dataset provided by scikit-learn. It contains 442 samples and 10 input features. Documentation: https://scikit-learn.org/stable/datasets/toy_dataset.html#diabetes-dataset

This dataset is used only for the course's introductory Linear Regression exercise. It is not the Smart Parking sensor dataset and does not demonstrate parking-classification performance. The project-specific classification dataset will be collected from the ESP32 and IR sensors during the hardware phase.

11. Expected Outcomes

Expected deliverables include a prototype that detects entry and exit events, estimates occupancy, displays parking status, and demonstrates a servo barrier; a labeled CSV dataset with at least 500 records; and an evaluation comparing classification models. The final discussion will address sensor reliability, data quality, model performance, and the limitation of estimating total occupancy from entrance events.

12. References

Idris, M. Y. I., Leng, Y. Y., Tamil, E. M., Noor, N. M., & Razak, Z. (2009). Car park system: A review of smart parking system and its technology. Information Technology Journal, 8(2), 101–113.

Geng, Y., & Cassandras, C. G. (2013). New smart parking system based on resource allocation and reservations. IEEE Transactions on Intelligent Transportation Systems, 14(3), 1129–1139.

Khanna, A., & Anand, R. (2016). IoT based smart parking system. In 2016 International Conference on Internet of Things and Applications (IOTA).

Scikit-learn developers. Diabetes dataset documentation. https://scikit-learn.org/stable/datasets/toy_dataset.html#diabetes-dataset

Before submission: verify the three research references, confirm Arduino IDE installation, replace estimated prices with local quotations, and update the proposal with actual hardware and data-collection details after testing.